Technology stack

English version of ../STACK.md. The document pins the versions and the purpose of the RTK CRM components (case 6, LCT 2026).

Backend

Component	Version	Purpose
Python	3.11	Language and runtime (`python:3.11-slim` image)
FastAPI	0.141.1	HTTP API, validation, OpenAPI
Uvicorn	0.34.0	ASGI server
Pydantic / pydantic-settings	2.10.4 / 2.7.1	Schemas and configuration from the environment
SQLAlchemy (async)	2.0.36	ORM, async engine
asyncpg	0.30.0	PostgreSQL driver
aiosqlite	0.20.0	SQLite driver for dev and tests
Alembic	1.14.0	Schema migrations
python-jose, passlib[bcrypt]	3.5.0 / 1.7.4	JWT and password hashing
Redis client	5.2.1	Report cache, potentially rate-limit counters
minio	7.2.10	S3-compatible file storage
APScheduler	3.11.0	Background checks for "stalled" requests
prometheus-fastapi-instrumentator	8.1.0	Metrics on `/metrics`
openpyxl, xlwt, xlrd, reportlab	—	Report export XLSX/XLS/PDF
loguru	0.7.3	Logging
pytest, pytest-asyncio	9.1.1 / 1.4.0	Tests

Frontend

Component	Version	Purpose
React	18.2	UI
TypeScript	5.2	Typing
Vite	5.0	Build and dev server
Ant Design	5.12	Component library
React Router	6.20	Routing
TanStack Query	5.x	Server state and cache
Zustand	4.4	Client state
Axios	1.6	HTTP client
Recharts	3.x	Report charts
dnd-kit	6.x / 8.x	Drag-and-drop of funnel stages

Infrastructure

Component	Purpose
PostgreSQL	Main database (Yandex Managed PostgreSQL, a container on the VM, or Supabase)
Redis / KeyDB	Report aggregate cache; KeyDB is recommended by the customer as lighter
S3 storage	Request files: Yandex Object Storage or MinIO (S3-compatible API)
Keycloak	Groundwork for corporate authentication (a closed contour)
Nginx	Reverse proxy, TLS, static file serving
Docker Compose	Starting the whole contour with one command

External services and integrations

- **Yandex Cloud** — the main contour: Compute (VM), Managed PostgreSQL, Object Storage, Container Registry, Cloud DNS.
- **Render + Vercel + Supabase** — an alternative contour without infrastructure of its own (API hosting, frontend static files, PostgreSQL).
- **GigaChat** — a summary for an interaction card (FT-6). Connected with an authorization key; without it the feature is off and the application works as usual.
- **LMS/Laravel** — integration groundwork (httpx), not in the MVP.

Environment requirements

- Python 3.11+, Node.js 20+.
- For the full contour: Docker and Docker Compose.
- The minimum for running the API without infrastructure: SQLite + a local `uploads/`.